

Relative Binding Affinity of Anabolic-Androgenic Steroids: Comparison of the Binding to the Androgen Receptors in Skeletal Muscle and in Prostate, as well as to Sex Hormone-Binding Globulin*

TÖNU SAARTOK, ERIK DAHLBERG,† AND JAN-ÅKE GUSTAFSSON

Department of Medical Nutrition, Karolinska Institutet, Huddinge University Hospital F 69, S-141 86 Huddinge, Sweden

ABSTRACT. It is unclear whether anabolic steroids act on skeletal muscle via the androgen receptor (AR) in this tissue, or whether there is a separate anabolic receptor. When several anabolic steroids were tested as competitors for the binding of [³H]methyltrienolone (MT; 17 β -hydroxy-17 α -methyl-4,9,11-estratrien-3-one) to the AR in rat and rabbit skeletal muscle and rat prostate, respectively, MT itself was the most efficient competitor. 1 α -Methyl-5 α -dihydrotestosterone (1 α -methyl-DHT; mesterolone) bound most avidly to sex hormone-binding globulin (SHBG) [relative binding affinity (RBA) about 4 times that of DHT]. Some anabolic-androgenic steroids bound strongly to the AR in skeletal muscle and prostate [RBAs relative to that of MT: MT > 19-nortestosterone (NorT; nandrolone) > methenolone (17 β -hydroxy-1-methyl-5 α -androst-1-en-3-one) > testosterone (T) > 1 α -methyl-DHT]. In other cases, AR binding was weak (RBA values < 0.05): stanozolol (17 α -methyl-5 α -androstano[3,2-c]pyrazol-17 β -ol), methanedieneone (17 β -hydroxy-17 α -methyl-1,4-androstadien-3-one), and fluoxymesterolone (9 α -fluoro-11 β -hydroxy-17 α -methyl-T). Other compounds had RBAs too low to be determined (*e.g.* oxymetholone (17 β -hydroxy-2-hydroxymethylene-17 α -methyl-5 α -androst-3-one)

and ethylestrenol (17 α -ethyl-4-estren-17 β -ol). The competition pattern was similar in muscle and prostate, except for a higher RBA of DHT in the prostate. The low RBA of DHT in muscle was probably due to the previously reported rapid reduction of its 3-keto function to metabolites, which did not bind to the AR [5 α -androstane-3 α ,17 β -diol and its 3 β -isomer (3 α - and 3 β -adiol, respectively)]. Some anabolic-androgenic steroids (only a few synthetic) bound to SHBG (1 α -methyl-DHT >> DHT > T > 3 β -adiol > 3 α -adiol = 17 α -methyl-T > methenolone > methanedieneone > stanozolol). The ratio of the RBA in rat muscle to that in the prostate (an estimate of the myotrophic potency of the compounds) was close to unity, varying only between about 0.4 and 1.7 in most cases. The present data indicate that 1) the existence of a putative anabolic receptor distinct from the AR must be questioned, 2) many anabolic steroids interact with the AR (generally with lower RBA than NorT or T), 3) some steroids with anabolic-androgenic activity *in vivo* do not bind to the AR, and must have an indirect mechanism of action (*e.g.* via biotransformation to active compounds, by influencing the metabolism of other steroids, or by displacing them from SHBG). (*Endocrinology* 114: 2100, 1984)

IT IS WIDELY held that direct steroid hormone action is exerted via receptors specific for each class of hormonal steroids (1, 2). This implies, *e.g.* that hormone-receptor complexes must be formed in order for the steroids to act on the target cells. Skeletal muscle from rats (3–8), mice (6),¹ pigs (9, 10), rabbits (11), and man (12, 13), as well as cultured muscle cells (14), contain receptors for androgens and glucocorticoids. In the case of rats and rabbits, an estrogen receptor has also been

found (11, 15). All three receptors are present in the rat levator ani muscle as well (see Refs. 3–15), and the androgen receptor (AR) in this muscle can be translocated *in vitro* into isolated cardiac muscle cell nuclei (16). The physicochemical properties of these receptors are very similar to those of receptors in the classical target tissues. Furthermore, the receptors in muscle tissue are subject to endocrine regulation. These data speak in favor of a physiological role of the steroid hormone receptors in skeletal muscle.

Androgens, in addition to their androgenic action on male reproductive tissues such as the prostate, have a protein-anabolic action in skeletal muscle (17–21). Anabolic-androgenic steroids have been developed with the aim of retaining the protein-anabolic action on muscle while reducing the androgenic effects of the steroids (17–21). The question whether myotrophic and androgenic actions of steroids can be dissociated deserves further

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† To whom correspondence and requests for reprints should be addressed.

¹ Nyberg, L., E. Dahlberg, K. Lundström, P. Jonsson, and L.-E. Edqvist, submitted for publication.

attention in view of a number of considerations. The relative anabolic potency has been assessed by various biological assays (see *e.g.* Refs. 17, 21, 22), *e.g.* comparing a myotrophic effect (weight increase of the levator ani muscle) with an androgenic effect (weight increase of accessory sex organs) in castrated male rats (23, 24). For various reasons (*cf.* below), the levator ani is not a typical skeletal muscle, and the validity of the assay has been seriously questioned (see Refs. 17, 21, 22). The rat levator ani muscle is in fact the dorsal bulbocavernosus muscle, which is peculiar to rodents, and which does not correspond to the levator ani muscle in man, cat, or dog (25). It is trophically dependent (not only sensitive) on androgens (26–31), and its androgen receptor concentration is intermediate to those of accessory sex tissues and skeletal muscle, respectively (see *e.g.* Ref. 32). Moreover, very few studies on anabolic-androgenic steroids have focused on the interaction with receptors (especially in skeletal muscle), but rather on nitrogen retention. It is also unclear whether an anabolic receptor distinct from the AR exists or not. Despite lacking information as to whether they have any strength-promoting effects, anabolic-androgenic steroids have been banned by sporting associations. Some of these steroids (notably 17 α -methylated compounds) may *e.g.* cause a variety of hepatic lesions (33, 34). Irrespective of these considerations, they are widely misused by athletes (34–37), including sporting children (38). These circumstances call for investigations into the possible mechanism of action of anabolic-androgenic steroids, especially in skeletal muscle.

To study whether anabolic-androgenic steroids can act via the AR(s) in skeletal muscle and prostate (*i.e.* whether they have the capacity to bind), and (if so) to compare their relative binding affinities (RBAs) to these receptors, we carried out the present investigation. We also wanted to study the serum binding of the compounds, since steroid-binding proteins in serum must be assumed to affect the availability of these steroids to the target tissues.

Materials and Methods

Chemicals

[17 α -Methyl-³H]MT² (SA, 87.0 Ci/mmol), [1,2-³H]dihydrotestosterone (DHT) (SA, 51.6 Ci/mmol), [³H]toluene (SA, 2.73

$\times 10^6$ dpm/ml), and unlabeled MT, were purchased from New England Nuclear Corp. (Boston, MA). The radioactive steroids were repeatedly purified by silica-gel thin layer chromatography to keep the radiopurity above 99%, since radioactive contaminants may cause serious interferences in receptor research (39). Testosterone (T), DHT, 5 α -androstan-3 α ,17 β -diol, 5 α -androstan-3 β ,17 β -diol, and fluoxymesterone were purchased from Sigma Chemical Co. (St. Louis, MO). 17 α -Methyltestosterone, 17 α -hydroxy-5 α -androstan-3-one, 17 α -hydroxy-4-androstan-3-one, and nandrolone (19-norT) were purchased from Steraloids Inc. (Wilton, NH). Mesterolone (1 α -methyl-DHT) and methenolone were kindly provided by Schering AG (Berlin/Bergkamen, West Germany), methanedieneone by Ciba-Geigy AG (Basel, Switzerland), ethylestrenol by N. V. Organon (Oss, The Netherlands), stanozolol by Sterling-Winthrop Drug Inc. (New York, NY), and oxymetholone by Syntex Pharmaceuticals Ltd. (Maidenhead, Berkshire, UK). Insta-Gel and Lipidex 1000 were products of Packard Instrument Co. (Downers Grove, IL), and Dextran T 70 was from Pharmacia Fine Chemicals (Uppsala, Sweden). Charcoal (Norit A, neutralized) was purchased from Sigma, and was treated as described elsewhere (6) before use. All other chemicals were reagent grade products from Sigma or from Merck A. G. (Darmstadt, West Germany).

Animals and tissues and preparation of cytosol

Ten-week-old Sprague-Dawley rats of both sexes (Anticimex, Stockholm, Sweden) were used to prepare cytosol from skeletal muscle (pooled from 25 females) and prostate (pooled from 25 males). New Zealand White rabbits of both sexes (Axell's Animal Farm, Näsby, Sweden), weighing 1.5–2.5 kg at the age of 2–5 months, were also used to prepare skeletal muscle cytosol. The animals were killed by concussion followed by cervical dislocation. Muscles (see Ref. 11) were resected, cleaned from visible fat and connective tissue, and frozen in liquid nitrogen within 5–10 min. Rabbit muscle was stored at –70 C for about a month and rat muscle was analyzed the day after resection/freezing. Prostatic cytosol was prepared from fresh tissue (*cf.* below). To obtain human serum, venous blood was collected from a healthy 30-yr-old male, and the serum was stored frozen (–70 C) for 1 day before the analysis. For comparison we also used serum from a 30-yr-old female pregnant in the third trimester.

Muscle cytosol was prepared by grinding the frozen tissue to a powder in liquid nitrogen and mixing with 1 vol buffer [50 mM Tris-HCl, pH 7.4 (23 C), 1 mM EDTA, 0.1 mM 2,3-dihydroxy-1,4-dithiol-butane, 10% (vol/vol) glycerol] to a 1:1 (wt/vol) homogenate (6, 11). Prostates were homogenized in 3 vol buffer using an Ultra-Turrax homogenizer (6). A 105,000 $\times g_{av}$ supernatant was used as cytosol. Prostatic cytosol and human serum were passed through columns of Lipidex 1000 at 4 and 22 C, respectively, in order to reduce the concentration of endogenous steroids before incubation (40).

Ligand specificity studies

Incubations of cytosols from rat (5, 6) and rabbit (11) muscle and of rat prostate (41) were performed as described, as were those of human serum (40). All incubations were terminated

² The following abbreviations and trivial names are used: methyltrienolone (MT), 17 β -hydroxy-17 α -methyl-4,9,11-estratrien-3-one; 5 α -dihydrotestosterone (DHT), 17 β -hydroxy-5 α -androstan-3-one; nandrolone [19-nortestosterone (19-norT)], 17 β -hydroxy-4-estren-3-one; 1 α -methyl-DHT (mesterolone), 17 β -hydroxy-1 α -methyl-5 α -androstan-3-one; fluoxymesterone, 9 α -fluoro-11 β -hydroxy-17 α -methyltestosterone (methyl-T); oxymetholone, 17 β -hydroxy-2-hydroxymethylene-17 α -methyl-5 α -androstan-3-one; ethylestrenol, 17 α -ethyl-4-estren-17 β -ol; methanedieneone, 17 β -hydroxy-17 α -methyl-1,4-androstadien-3-one; methenolone, 17 β -hydroxy-1-methyl-5 α -androstan-1-en-3-one; stanozolol, 17 α -methyl-5 α -androstan-3-one-17 β -ol; SHBG, sex hormone-binding globulin (testosterone-binding globulin); RBA, relative binding affinity.

by adsorption of unbound steroids to dextran-coated charcoal (5, 6, 40, 41). The final concentrations of [^3H]MT were selected on the basis of previous saturation analyses, and varied between 0.7 and 0.9 nM for all cytosols. Male and female human serum was diluted 8-fold and 100-fold, respectively, at which dilutions at 1.3 nM concentration of ^3H -labeled DHT just saturated the DHT-binding capacity in the serum samples used (determined in a separate experiment). In another experiment (not shown) it was verified that DHT was not metabolized during the serum incubations. Under these conditions (40) DHT is mainly sex hormone-binding globulin (SHBG)-bound (42, 43). All competitors (Table 1) were added at eight concentrations representing approximately 1- to 128-fold molar excess compared with the ^3H -labeled ligand, with 2-fold differences between the concentrations (41). Nonspecific binding was determined at a 1000-fold excess of the unlabeled ligand and the highest concentration of each competitor (5, 41). Radioactivity was then determined after charcoal treatment as described elsewhere (44). The RBA was calculated from logit-log plots as the ratio between the molar concentrations of unlabeled ligand and of competitor required to displace 50% of the ^3H -labeled ligand from the specific binding sites (6, 45).

Results

Most anabolic-androgenic steroids on the Swedish drug market (1982) and some other steroids were tested

TABLE 1. RBAs of anabolic-androgenic steroids as competitors for human serum-protein binding of 5α -DHT (mainly SHBG), and for the receptor binding of methyltrienolone in cytosol from rabbit and rat skeletal muscle and from rat prostate

Competitor	Human serum (SHBG)	Rabbit muscle	Rat muscle	Rat prostate	Ratio rat muscle-rat prostate
1. Methyltrienolone	<0.01	1	1	1	1
2. DHT	1	0.07	<0.01	0.46	0.03
3. 1α -Methyl-DHT	4.4	0.21	0.08	0.25	0.32
4. T	0.19	0.07	0.23	0.15	1.53
5. 19-NorT	0.01	0.20	0.24	0.60	0.40
6. 17α -Methyl-T	0.05	0.10	0.11	0.13	0.85
7. 5α -Androstane- $3\beta,17\beta$ -diol	0.17	ND	<0.01	0.01	0.46
8. 5β -Androstane- $3\alpha,17\beta$ -diol	0.05	ND	<0.01	0.01	0.08
9. 17α -Hydroxy-4-androsten-3-one	ND	ND	<0.01	<0.01	0.50
10. 5α -Androstan- 17α -ol-3-one	ND	ND	<<0.01	<<0.01	ND
11. Methenolone	0.03	0.09	0.24	0.14	1.67
12. Stanozolol	0.01	0.03	0.02	0.03	0.60
13. Methanediene	0.02	0.02	0.02	0.03	0.75
14. Fluoxymesterone	<0.01	0.02	0.01	0.02	0.77
15. Oxymetholone	<0.01	<0.01	<0.01	<0.01	1.54
16. Ethylestrenol	<0.01	0.01	<0.01	<0.01	2.00

ND, Not determined. The ratio is calculated from original data.

in an *in vitro* system as competitors for the AR. ^3H -Labeled MT was used as the ligand. The study was carried out both in rat and in rabbit skeletal muscle in order to estimate the myotrophic potency of the steroids as their RBA for the receptor. The androgenic potency was similarly estimated as the RBA of the compounds for the corresponding ligand-binding sites in the rat prostate. The capacity of the anabolic-androgenic steroids to compete with the binding of DHT to human serum was also estimated. This binding is mainly due to SHBG (42, 43). The RBA values of the anabolic-androgenic steroids for these proteins were calculated from logit-log plots, and are given in Table 1.

Of the steroids tested, the most efficient competitor for the androgen receptor was MT itself (RBA, 1.00 by definition). This was true in skeletal muscle from both species and in the rat prostate. MT was followed by 19-norT and T in all three tissues (RBA values in Table 1). DHT was highly active in the rat prostate (RBA, 0.46), but not in rabbit (RBA, 0.07) or rat (RBA, 0.01) skeletal muscle. The low affinity of DHT for the AR in skeletal muscle compared to that in the prostate has been reported previously, and is probably due to a rapid metabolism of DHT to 5α -androstane- $3\alpha,17\beta$ -diol and its 3β -isomer in muscle cytosol (3, 5, 6). As can be seen in Table 1, these metabolites did not bind significantly to the androgen receptor. Whereas 1α -methyl-DHT was an efficient competitor in rat prostate (RBA, 0.25) and rabbit muscle (RBA, 0.21), its RBA was slightly lower in rat skeletal muscle (RBA, 0.08). 17α -Methyl-T was an about equally efficient competitor in each case (RBA values of about 0.1; Table 1). Also methenolone belonged to this group of steroids with relatively high, and generally similar, affinity for the androgen receptor in all three tissues.

In other cases, androgen receptor binding was weak (RBA values between 0.02 and 0.05). Stanozolol and fluoxymesterolone belonged to this group of steroids, none of which showed any differences in affinity between the AR in muscle or prostate. Methanediene, which cannot be metabolized to its more potent 5α -reduced counterpart (46), also bound weakly to all binding proteins analyzed.

For other compounds the RBA for the AR was insignificant, *i.e.* it was too low to be accurately determined (RBA, 0.01 or less). In fact, this was characteristic for a large class of steroids, including *e.g.* the DHT metabolites 5α -androstane- $3\alpha,17\beta$ -diol and 5α -androstane- $3\beta,17\beta$ -diol, and the synthetic compounds oxymetholone and ethylestrenol. Also here, the receptors in muscle and prostate showed the same competition pattern.

Rat and rabbit skeletal muscle showed only very slight differences in AR ligand specificity. The small differences seen may be due to a species difference, but perhaps

also to the different time of tissue storage. Moreover, the similarities between the ligand specificities of the receptor in skeletal muscle from the two species and that in rat prostate were considerably greater than the differences. Only in case of DHT was there a marked difference (*cf.* above).

The ligand specificity of SHBG, however, was markedly different from that of the receptor(s). Apart from 1 α -methyl-DHT (mesterolone), which was an extraordinary potent competitor for SHBG (RBA about 4 times that of DHT), DHT itself was the most efficient competitor (RBA, 1.00 by definition), followed by T (RBA, 0.19). Both 5 α -androstane-3 β ,17 β -diol, and its 3 α -isomer bound to SHBG (RBA values of 0.17 and 0.05, respectively) but not to the receptor(s) (RBA values of 0.01 or less in all cases). 17 α -Methyl-T bound to SHBG with the same affinity as 5 α -androstane-3 α ,17 β -diol, whereas methenolone, methanediene, and stanozolol were weaker competitors. Some steroids which lacked affinity for the AR were also unable to bind significantly to SHBG, *e.g.* fluoxymesterolone, oxymetholone, and ethylestrenol. MT, despite being the strongest competitor for all receptor-containing tissues, did not bind at all to SHBG.

To verify that the fact that we used male serum, and not the classical system containing high SHBG levels, *i.e.* human female plasma from the third trimester of pregnancy, we made some additional experiments (Table 2). We compared the binding of some steroids to male plasma and third trimester plasma, both before and after delipidation by a method efficient in reducing steroid levels without destroying proteins (40). No significant differences were noted.

We have also made sure, using thin layer chromatog-

TABLE 2. Comparison between male serum and female serum from the third trimester of pregnancy, in some cases both before and after delipidation

Competitor	RBA in male plasma		RBA in third trimester plasma	
	Untreated	Delipidated	Untreated	Delipidated
DHT	1	1	1	1
1 α -Methyl-DHT	3.26	3.16	3.35	NA
T	0.03	0.04	0.04	0.04
19-NorT	0.01	0.01	NA	NA
Stanozolol	0.02	0.04	NA	NA
Estradiol	0.01	0.01	0.01	NA
Cortisol	0.01	0.01	NA	NA

The delipidation was carried out as described elsewhere (40) by passing plasma through a column of Lipidex 1000. The experiments before and after Lipidex were carried out using 100- and 50-fold (third trimester) and 4- and 2-fold (male) diluted plasma samples, to account for the dilution during gel filtration. Otherwise the experiments were accomplished as in *Materials and Methods*. NA, Not analyzed.

raphy on silica gel, that [3 H]DHT was not metabolized during the experiments (not shown). As to the above experiments using MT as the ligand, it is well established that this steroid is not metabolized in either muscle or prostate (see 3, 5, 6, 9, 15 for data and Refs.).

Other control experiments have indicated that although MT binds to the glucocorticoid receptor in muscle (3, 5, 6), AR estimations are unaffected by the presence of an excess of unlabeled dexamethasone (6), and in this study we have not seen any difference between RBA estimations for the AR carried out in the presence or absence of unlabeled dexamethasone (not shown).

The ratio of the RBA in rat muscle to that in the rat prostate was calculated as an estimate of the myotrophic potency of the compounds. This ratio was close to unity, varying between about 0.4 and 1.7 in most cases (Table 1).

Discussion

The concept of receptors has influenced the thinking in the field of steroid hormones so strongly that very few studies have taken into account the importance of high affinity plasma steroid-binding proteins in the regulation of the availability of the steroids to the target cells, and studied both types of proteins simultaneously (10).¹ The present study addresses this question in relation to anabolic-androgenic steroids. It would seem to be of little physiological significance that a steroid has a high affinity for its receptor if it does not become available to the target cell, or is rapidly metabolized in the tissue, as shown *e.g.* for DHT in muscle (3, 5).

The results in the present study are in agreement with published data on the interaction of some of the steroids with the rat prostatic AR (47, 48) and with the AR in rat skeletal muscle (3, 5, 6). The data support the contention (5, 6) that these proteins represent the same entity. As far as data are available, our results on SHBG-binding affinities are comparable to those previously reported (42, 43).

The apparent low affinity of DHT for the AR in muscle compared with that in prostate is in agreement with our previous data (3, 5) and is an effect of a rapid metabolism of DHT in muscle cytosol (3, 5) to metabolites inactive as competitors for the receptor, which yet bind to SHBG. It is possible that also 1 α -methyl-DHT is inactivated by a similar metabolism in muscle. This would explain the efficiency of this steroid as a competitor in serum and prostate, as opposed to its low affinity for the receptor in rat muscle. Yet, bearing the unusually high affinity of 1 α -methyl-DHT for SHBG in mind, it would seem less likely that this steroid will reach the target tissues in sufficient quantities to act directly in species which (unlike rodents) have a circulating SHBG. However, based

on the present data, 1α -methyl-DHT should be capable of increasing the free fraction of other SHBG-bound steroids to a considerable degree. Thus, although this steroid also has an intrinsic activity, such a displacement could provide a satisfactory explanation for its biological activity even if the compound did not reach its target cell.

Skeletal muscle differs from prostate in being virtually incapable of carrying out 5α -reduction, a main metabolic pathway for T in the prostate (for Refs., see *e.g.* 5, 48). Thus, muscle has a T/DHT ratio about an order of magnitude higher than that of prostate (49, 50). This ratio also differs between muscle and prostate as an effect of the fact that the 3-keto group of DHT is more readily reduced to inactive compounds in skeletal muscle (5) than in the prostate. The RBA of DHT was higher than that of T in the prostate (Table 1). However, since DHT is rapidly inactivated in muscle, T is conceivably a more important receptor ligand than DHT in muscle also for this reason. Increasing the T/DHT ratio *in vivo* by a 5α -reductase inhibitor (51) should affect muscle if this is correct. Whereas 5α -reduction of T and 19-norT is about equally efficient, DHT has a considerably higher affinity for the AR than 5α -dihydro-19-norT (52, 53). Exogenous 19-norT would therefore be inactivated in the prostate, but not in skeletal muscle. In view of its higher RBA, 19-norT can be expected to be an even more potent anabolic agent than T in muscle.

The data in the present paper strongly argue against the possibility of an anabolic receptor. On the contrary, they support the contention (5, 6) that skeletal muscle contains an AR very similar or identical to that in secondary sex organs. Our earlier studies on physicochemical properties (5), endocrine regulation (6), and ligand specificity (5, 6) of the AR also failed to demonstrate any differences between the receptor in skeletal muscle and the AR in classical target tissues.

The anabolic-androgenic ratio was calculated as the RBA in rat muscle divided by that in the rat prostate, and varied between about 0.4 and 1.7 for all but three compounds. This was a fact irrespective of RBA values. This ratio is considerably less than that reported for the ratio between the *in vivo* activity of these steroids in the levator ani and in sex tissues (seminal vesicle or prostate) (*e.g.* Ref. 21). Thus, the *in vivo* anabolic-androgenic ratio varies in different studies from 0.4 to 1.9 for DHT, from 1.5 to 15 for 19-norT, from 1.8 to 10.6 for oxymetholone, and from 1.4 to 10.6 for stanozolol (see Ref. 21). This variability of *in vivo* data, and the discrepancy between our *in vitro* data and these *in vivo* potency estimates, are not surprising in view of the many factors that may influence *in vivo* activity, *e.g.* peroral availability, hepatic metabolism, distribution into adipose tissue, possible indirect effects *etc.* In line with this, we recently com-

pared the relationship between receptor affinity and *in vivo* activity of glucocorticoids, and found that whereas a topical edema-inhibition test on rat skin correlated well with receptor affinity ($r = 0.98$), thymus involution after sc or peroral steroid administration did not (8).

Some interesting differences between binding of ligands to SHBG and the AR were found. Removal of the 19-methyl group of T increased the RBA for the receptor in rat prostate and in rabbit (but not rat) muscle 4- and 3-fold, respectively, whereas the RBA for SHBG was reduced 19-fold. The introduction of a 17α -methyl group into T negatively influenced the RBA for SHBG, but had only a minor influence on the RBA for the AR. Whereas 1α -methylation of DHT reduced the RBA for the rat prostatic receptor, the affinity for the receptor in muscle from both species increased. Most remarkably, however, we found that 1α -methyl-DHT was about 4 times more active as a ligand for SHBG than DHT was. A 1-ene double bond may have diverse effects depending on the structure of the parental steroid. The introduction of a 1-ene double bond into 17α -methyl-T (giving methanediene) reduced the RBA for both SHBG and the androgen receptor. On the other hand, comparing methenolone with 1α -methyl-DHT reveals a different effect of 1-unsaturation in the presence of a 1-methyl group. In this instance, a 1-ene structure resulted in an about 2-fold lowered RBA for the AR in rabbit muscle and rat prostate, an about 2-fold increased affinity for the receptor in rat muscle, but an at least 30-fold reduced RBA for human SHBG. It is difficult to explain why the effect of certain substitutions depends on structure of the parental steroid. This is not peculiar to the 1-ene double bond in the present study, however; we have recently obtained similar results for 9α -fluorination of different glucocorticoids (8).

We have previously estimated the RBA in muscle for some other steroids with an anabolic action *in vivo* (6). Neither the 11β -hydroxy group, nor the presence of a 1,4-diene A-ring, in several of these compounds (6) can explain their lack of interaction with the androgen receptor in view of the facts that fluoxymesterolone (11β -hydroxy group), methenolone (1-ene), and methanediene (1,4-diene) all bound (albeit weakly) to the AR in the present study. However, several of these anabolics carried different substituents at C-2 (6), and in the present study we found that oxymetholone was unable to compete with the AR. Since the basic structure of oxymetholone, apart from its C-2 substituent, is 17α -methyl-DHT, which should be capable of interacting with the AR, the C-2 substitutions (6) probably inhibit binding to AR.

Based on the present data, the steroids with anabolic-androgenic potency *in vivo* may be divided into four groups, *viz.* 1) those that bind well to the receptor (both

in muscle and prostate) but inefficiently to SHBG, 2) those that bind considerably better to SHBG than to the receptor, 3) those that bind about equally well both to SHBG and to the receptor, and 4) those that lack the capacity of binding to either protein.

The mechanism of action for these groups of anabolic-androgenic steroids may well be receptor mediated for all but the last of these groups. It seems reasonable to assume that different metabolism of these hormones in muscle and prostate will influence their activity. Also, in particular for groups 2) and 3), both the affinity of the steroids for SHBG and the concentration of this protein will affect the *in vivo* activity of the steroids. Group 4) is interesting in view of the discrepancy between *in vivo* and *in vitro* activity in this case. Thus, a different mechanism of action must be postulated for these steroids. Possibly, they might influence enzymatic activities involved in metabolic pathways regulating protein anabolism in muscle, e.g. the metabolism of anabolic or catabolic steroids.

In conclusion, we have provided evidence that some anabolic-androgenic steroids interact with the AR in rabbit or rat skeletal muscle, as well as with the AR in the rat prostate. Other representatives of this class of steroids do not interact with the AR, and some alternative mechanisms of action of these compounds have been suggested. In general the ligand specificity of the AR was the same in muscle and prostate. Hence, the existence of a specific anabolic receptor distinct from the AR must be seriously questioned. In view of current knowledge of steroid hormone action via receptor interaction, the present data may be taken as an indication that some of the anabolic steroid hormones have the potential to act directly on skeletal muscle.

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